nature portfolio

Peer Review File

Integrative Analysis of H&E and IHC Identifies Prognostic Immune Subtypes in HPV Related Oropharyngeal Cancer

This manuscript has been previously reviewed at another Nature Portfolio journal. This document only contains reviewer comments and rebuttal letters for versions considered at Communications Medicine.

Reviewers' comments:

Reviewer #1 (Remarks to the Author):

In their study titled "Integrative Spatial Analysis of H&E and IHC Images Identifies Prognostic Immune Subtypes Correlated with Progression-Free Survival in Human Papillomavirus (HPV)-Related Oropharyngeal Squamous Cell Carcinoma", Sumanth Reddy Nakkireddy et al. analyzed cases of OPSCC using deep learning, advanced image registration, and correlated their findings with prognosis in a cohort of patients with relapsed/non-relapsed disease. While the study appears timely and intriguing, significant language issues render parts of the manuscript challenging to comprehend in detail. Considering the extensive literature on TILs in cancer, particularly in OPSCC – with or without the use of deep learning – the authors should consider including these studies for comparison and to rationalize their study's contributions. Overall, the study's sample size is relatively small, and although the authors attempt to demonstrate the prognostic significance of various thresholds, a validation cohort – albeit challenging to obtain – would be beneficial.

Major Points:
• Additional literature on TILs in OPSCC and HPV-associated diseases, such as the quantification of TILs and the broader topic of TILs in cancer, should be discussed (please see below).
• Numerous language errors throughout the manuscript obscure a clear line of thought (examples provided below).
• The reproducibility of the study is questionable, given the unavailability of the dataset or the algorithm, possibly including weights. The authors must clarify what is released as code and what is considered "novel". Some links in the GitHub repository appear non-functional (notably, the crucial ones). The crucial method of image registration in the methods section is overly generic and could be enhanced with more details (note: this includes functional GitHub links).
• The model employed for tumor segmentation, originally developed for colorectal cancer, lacks clear performance evaluation on OPSCC cases.
• A detailed graphical representation of the patient population, including inclusion and exclusion criteria, sub-cohorts, and controls, is necessary. This also pertains to the sub-cohorts of immunological patterns later discussed.

Minor Points:
• Methodology: The rationale behind using the Cox proportional hazards model for calculating "Hazard ratios (HR) and 95% confidence intervals (CIs)" and a log-rank test for assessing differences in PFS should be clarified.
• Conclusion: The statement, "This approach offers the capability to leverage information from TME to stratify patients more effectively into subgroups with distinct survival outcomes, thereby advancing the field of precision oncology care for patients with HPV+ OPSCC," needs further elaboration. Specifically, how this retrospective study with a limited cohort, demonstrating a correlation of a biomarker to outcome, advances precision oncology care.

Abstract (examples of language issues, just randomly starting at the beginning)
• HPV(+)/ HPV (+) would probably be Human papilloma positive (HPV-positive)
• "(…) and 11 adjacent immunohistochemistry (IHC)-stained slides to investigate the cellular composition and functional characteristics across different regions within the TME (…)" 11 slides with 11 markers; difficult to understand if a reader browses the abstract?
• "(…) with high tumor-infiltrating lymphocytes (TIL) (…)" high levels of

Literature
• Almangush, A., Jouhi, L., Atula, T. et al. Tumour-infiltrating lymphocytes in oropharyngeal cancer: a validation study according to the criteria of the International Immuno-Oncology Biomarker Working Group. Br J Cancer 126, 1589–1594 (2022). https://doi.org/10.1038/s41416-022-01708-7
• Li, Q., Liu, X., Wang, D. et al. Prognostic value of tertiary lymphoid structure and tumour infiltrating lymphocytes in oral squamous cell carcinoma. Int J Oral Sci 12, 24 (2020). https://doi.org/10.1038/s41368-020-00092-3
• Corredor G, Toro P, Koyuncu C, Lu C, Buzzy C, Bera K, Fu P, Mehrad M, Ely KA, Mokhtari M, Yang K, Chute D, Adelstein DJ, Thompson LDR, Bishop JA, Faraji F, Thorstad W, Castro P, Sandulache V,

Koyfman SA, Lewis JS, Madabhushi A. An Imaging Biomarker of Tumor-Infiltrating Lymphocytes to Risk-Stratify Patients With HPV-Associated Oropharynageal Cancer. J Natl Cancer Inst. 2022 Apr 11;114(4):609-617. doi: 10.1093/jnci/djab215. PMID: 34850048; PMCID: PMC9002277.

Reviewer #2 (Remarks to the Author):

The current manuscript focuses on HPV+OPSCC and investigates the prognostic value of H&E and immune focused IHC based analysis. This is a worthwhile endeavor since further stratification of this entity is critical to development of precision oncology approaches.
There are several major methodological issues to start with.
1. Measuring TIL infiltration in immune tissue. The algorithm seems to based on the Park et al JCO 2022 algorithm which was designed for NSCLC. Because OPSCC is a squamous carcinoma occurring in a lymphoid background (tonsil, BOT), it is essential to separate true TILs from immunocytes surrounding the tumor and to demonstrate that the algorithm is trained to do that. As far as I can tell that was not done here so I am skeptical of the TIL measurements based on H&E. Several other teams have tried to plug pre-existing algorithms for TILs into OPSCC but this is a fundamental misapplication of an algorithm developed primarily on stromal tissue with cancer infiltration to a lymphoid tissue with squamous cancer infiltration; in my opinion.
2. The above limitation also applies to lymph nodes. It is essential to demonstrate that the algorithm is capable of separating TILs and only counting those inside the actual SCCA intra-nodal volume.
3. Presumably these are serial sections and if multiple IHC's were done that means the datasets are at least 3-5 cell nuclei apart in depth. How is misregistration handled when H&Es and IHCs are integrated and what is the misregistration error? Not quite sure how a pixel by pixel registration would be done considering these are going to differ by more than a few cell nuclei based on slide thickness alone……
4. There is no validation cohort. This is a fundamental limitation to the current study which is critically impactful to its validity. Because there is no validation cohort, the methodological limitations above are quite problematic. A validation cohort would at least provide a simple "but it works" response which would be at least partially sufficient translationally if not scientifically.
Followed by a general novelty issue.
1. This is known. There are now dozens of published papers on how TIL infiltration affects survival for OPSCC generally and HPV+OPSCC. It is quite troubling that none of those papers are actually cited or referenced here. Including papers on CD4+ and CD8+; IHC based papers and transcriptomics based papers.
2. It has also been shown that TIL infiltration and spatial organization is predictive of OS and DFS in HPV+OPSCC. In multiple independent patient cohorts. Again, troubling that that is neither cited nor referenced here since it precedes the current effort by some time now.

Based on these limitations I would not recommend the manuscript for publication in its current form.

Reviewers' comments:

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In their study titled "Integrative Spatial Analysis of H&E and IHC Images Identifies Prognostic Immune Subtypes Correlated with Progression-Free Survival in Human Papillomavirus (HPV)-Related Oropharyngeal Squamous Cell Carcinoma", Sumanth Reddy Nakkireddy et al. analyzed cases of OPSCC using deep learning, advanced image registration, and correlated their findings with prognosis in a cohort of patients with relapsed/non-relapsed disease. While the study appears timely and intriguing, significant language issues render parts of the manuscript challenging to comprehend in detail. Considering the extensive literature on TILs in cancer, particularly in OPSCC – with or without the use of deep learning – the authors should consider including these studies for comparison and to rationalize their study's contributions. Overall, the study's sample size is relatively small, and although the authors attempt to demonstrate the prognostic significance of various thresholds, a validation cohort – albeit challenging to obtain – would be beneficial.

Major Points:

1. Additional literature on TILs in OPSCC and HPV-associated diseases, such as the quantification of TILs and the broader topic of TILs in cancer, should be discussed (please see below).

   Response: Thank you for your valuable feedback. We appreciate your suggestion and have duly incorporated significant background literature to strengthen the justification for our work. Additionally, we have included relevant literature on TILS (Tumor-Infiltrating Lymphocytes) in colon, breast cancer, and other cancer types in our reference section.

2. Numerous language errors throughout the manuscript obscure a clear line of thought (examples provided below).

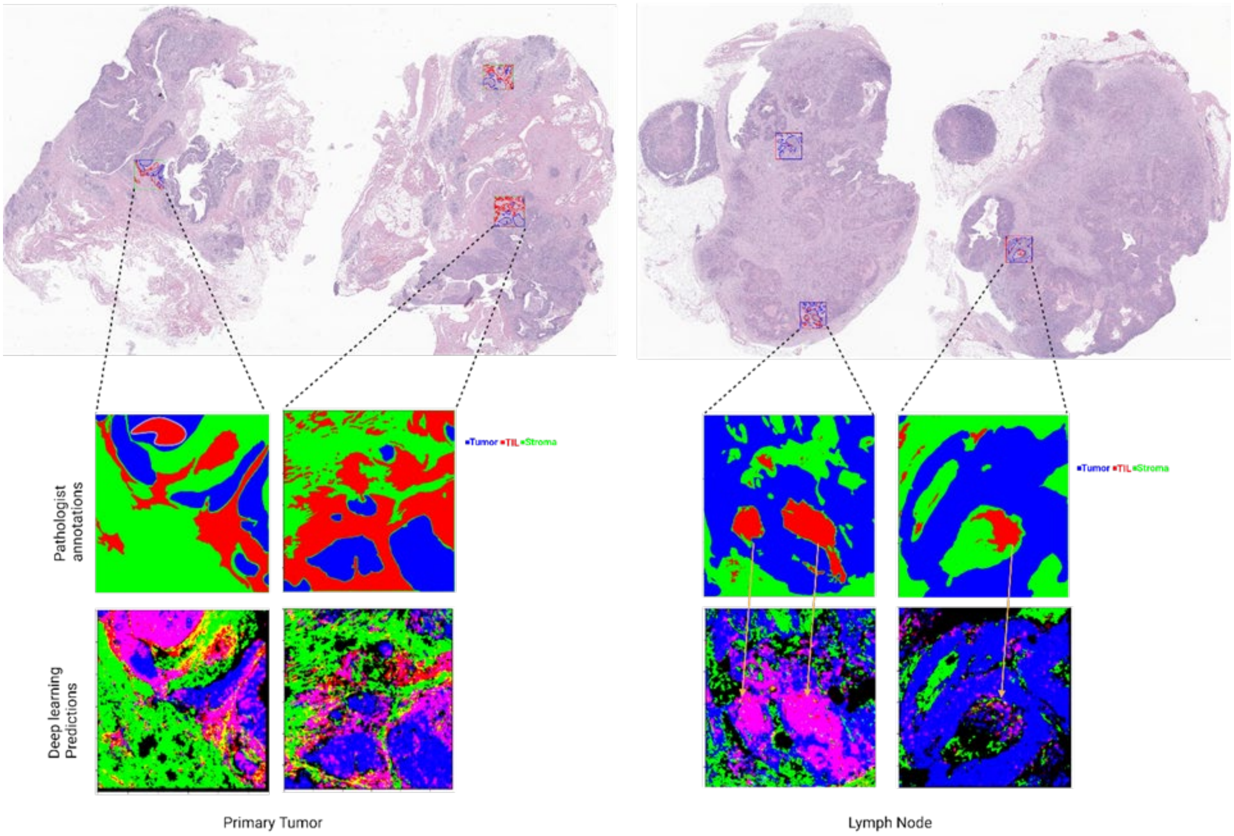
   Response: Thank you for the feedback, we have carefully corrected the manuscript.

• The reproducibility of the study is questionable, given the unavailability of the dataset or the algorithm, possibly including weights. The authors must clarify what is released as code and what is considered "novel". Some links in the GitHub repository appear non-functional (notably, the crucial ones). The crucial method of image registration in the methods section is overly generic and could be enhanced with more details (note: this includes functional GitHub links).

Response: We understand that without the data availability, it is hard to validate the results. Due to the patient's privacy, we cannot share the raw data of H&E and IHC images. However, we have uploaded all the downstream analysis data to the functional GitHub (https://github.com/hwanglab/HE_IHC_HN_analysis/blob/master/). Please let us know if you still have trouble accessing the link. All the codes used for producing the figures in the manuscript are made available in the previous link. The image registration method is made available to the public now, please follow the link https://github.com/hwanglab/WSI_registration.git, to access the code. Please email us back if you still have a problem accessing the codes.

• The model employed for tumor segmentation, originally developed for colorectal cancer, lacks clear performance evaluation on OPSCC cases.

Response: We have employed two pathologists with over 10 years of experience to identify the Tumor, TIL, and Stroma regions. The performance of the model is re-evaluated. The following figures show the concordance between the pathologist annotations and the Deep learning-based predictions that we have used in the manuscript.



• A detailed graphical representation of the patient population, including inclusion and exclusion criteria, sub-cohorts, and controls, is necessary. This also pertains to the sub-cohorts of immunological patterns later discussed.

Response: The graphical representation of the inclusion and exclusion criteria is added to the supplementary figures. Please check the following plot.

	CD163	CD20	CD3	CD4	CD45	CD8	ER Alpha	ER Beta	FoxP3	PD-L1	KRT AE1AE3	CD163	CD20	CD3	CD4	CD45	CD8	ER Alpha	ER Beta	FoxP3	PD-L1	KRT AE1AE3
P1	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O
P2	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O		O	O	O
P3	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O		O	O	O
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Minor Points:

• Methodology: The rationale behind using the Cox proportional hazards model for calculating "Hazard ratios (HR) and 95% confidence intervals (CIs)" and a log-rank test for assessing differences in PFS should be clarified.

Response: Thank you for your insightful feedback. Our study primarily aimed to investigate the relationship between image-based biomarkers and progression-free survival (PFS), while considering several other cofactors including sex, stage, treatment status, smoking status, and ACE score. To address this, we employed the Cox proportional hazards model, which proved to be an appropriate choice for conducting multivariate analysis. As illustrated in Figure 5, the results of our analysis highlight that image-based biomarkers emerge as the sole predictors of PFS, underscoring their significance in predicting disease progression and patient outcomes.

Furthermore, to provide additional evidence of the predictive power of these biomarkers, we conducted univariate Kaplan-Meier analysis, as depicted in Figure 4. This analysis further emphasizes the robustness of the image-based biomarkers in predicting progression-free survival independently. By employing both multivariate Cox proportional hazards modeling and univariate Kaplan-Meier analysis, we aimed to comprehensively evaluate the prognostic value of the image-based biomarkers in our study cohort.

• Conclusion: The statement, "This approach offers the capability to leverage information from TME to stratify patients more effectively into subgroups with distinct survival outcomes, thereby advancing the field of precision oncology care for patients with HPV+ OPSCC," needs further elaboration. Specifically, how this retrospective study with a limited cohort, demonstrating a correlation of a biomarker to outcome, advances precision oncology care.

Response: Thank you for your thoughtful review. We conducted this retrospective study to explore the potential of using image-based biomarkers to predict progression-free survival. While obtaining a validation group with similar data was challenging, our main aim was to identify these biomarkers that could effectively distinguish patients with longer progression-free survival. Our findings suggest that these biomarkers (patients with TILs high, CD8 high, and CD163 low) could help categorize patients into groups with either good or poor prognosis, which could lead to more careful consideration of treatment options. We believe our study provides valuable insights into the potential of using image-based biomarkers to improve patient care and treatment decisions.

Abstract (examples of language issues, just randomly starting at the beginning)

• HPV(+)/ HPV (+) would probably be Human papilloma-positive (HPV-positive)

• “(…) and 11 adjacent immunohistochemistry (IHC)-stained slides to investigate the cellular composition and functional characteristics across different regions within the TME (…)“ 11 slides with 11 markers; difficult to understand if a reader browses the abstract?

• "(...) with high tumor-infiltrating lymphocytes (TIL) (...)" high levels of

Response: Thank you for pointing out the issues. We have corrected these sentences in the main draft.

Literature

• Almangush, A., Jouhi, L., Atula, T. et al. Tumour-infiltrating lymphocytes in oropharyngeal cancer: a validation study according to the criteria of the International Immuno-Oncology Biomarker Working Group. Br J Cancer 126, 1589–1594 (2022). https://doi.org/10.1038/s41416-022-01708-7

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Response: Thank you for pointing out the existing literature work. We have added the references and edited the introduction in the main draft.

Reviewer #2 (Remarks to the Author):

The current manuscript focuses on HPV+OPSCC and investigates the prognostic value of H&E and immune focused IHC based analysis. This is a worthwhile endeavor since further stratification of this entity is critical to development of precision oncology approaches.

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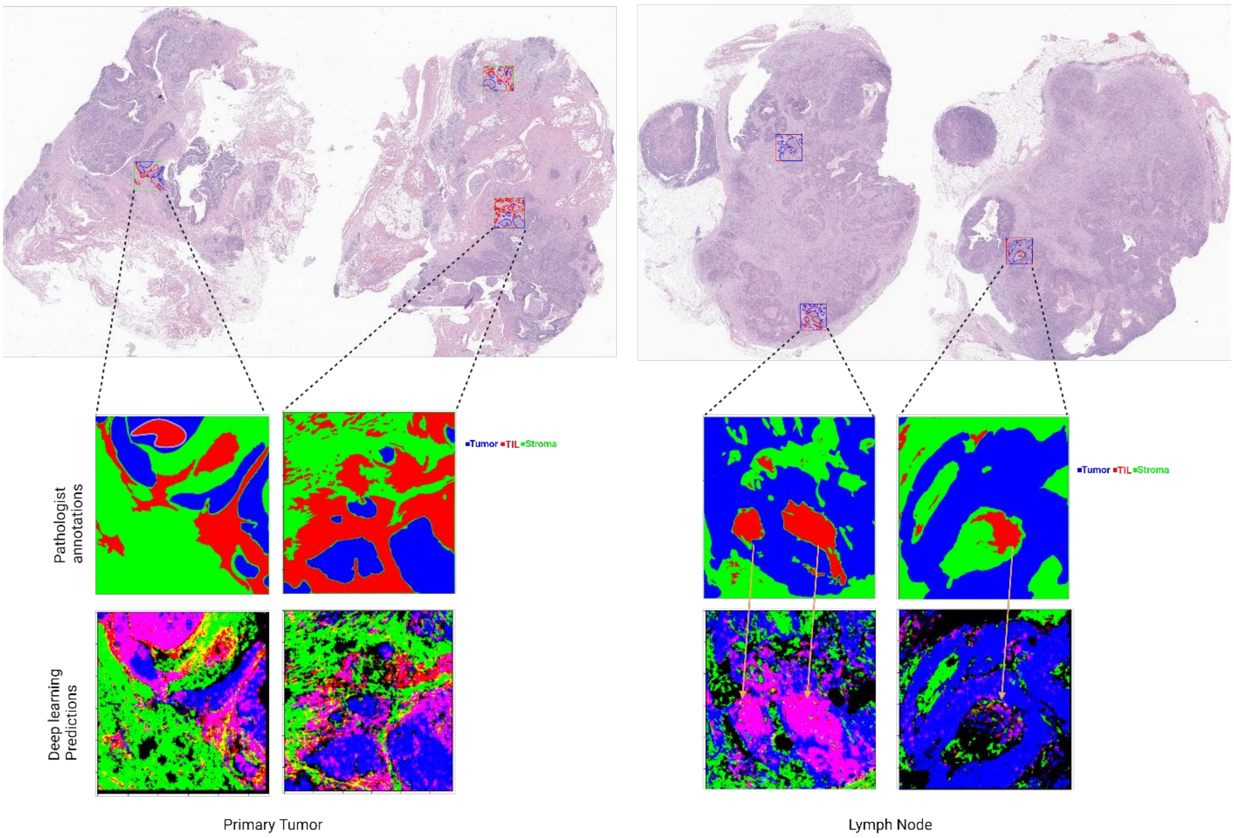
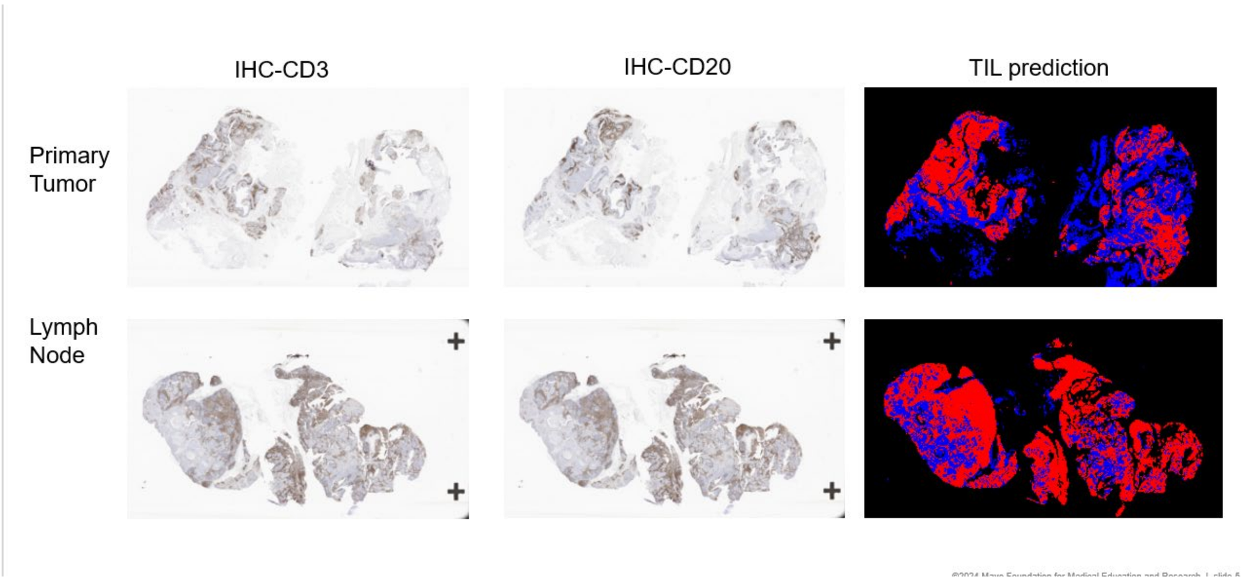
   Response: Thank you for your comment. In our study, we ensured robustness by involving two pathologists, each with a decade of experience, to accurately identify the tumor, tumor-infiltrating lymphocytes (TIL), and stroma regions. To validate the performance of our model, we conducted a re-evaluation process. We have included figures below illustrating the agreement between the annotations made by the pathologists and the predictions generated by our deep learning-based model. These figures demonstrate the concordance between the two methods, further reinforcing the reliability of our approach.



In addition to the previous comparisons, the figure below shows the overall concordance of the general immune cell population with the TILs predictions. The registered IHC slides shown below provide an overall spatial location of TILs on the H&E.



2. The above limitation also applies to lymph nodes. It is essential to demonstrate that the algorithm is capable of separating TILs and only counting those inside the actual SCCA intra-nodal volume.
   Response: Please refer to the figure attached in the previous response.

3. Presumably these are serial sections and if multiple IHC's were done that means the datasets are at least 3-5 cell nuclei apart in depth. How is misregistration handled when H&Es and IHCs are integrated and what is the misregistration error? Not quite sure how a pixel-by-pixel registration would be done considering these are going to differ by more than a few cell nuclei based on slide thickness alone……
   Response: Thank you for bringing up the misregistration concern. We've taken steps to address this in the main methods section of our manuscript. The purpose of registering H&E and IHC images is not to match cell to cell exactly but to determine which regions of the H&E are generally enriched with IHC phenotypes. Our registration approach involves employing a registration method based on pixel matching. Initially, we identified six to ten anchor pixels in both the immunohistochemistry (IHC) and corresponding hematoxylin and eosin (H&E) images. Subsequently, we utilized a rigid and non-rigid registration pixel-matching method, specifically ElasticX, to register the images. For further details, please refer to the GitHub repository: https://github.com/hwanglab/WSI_registration.git.
   Given that our tumor-infiltrating lymphocytes (TILs) scoring method operates at the tile-level prediction, it's important to note that the IHC quantification approximately aligns with the region of the tile due to inherent limitations of the registration method. Unfortunately, it was not possible to measure the numeric error for misalignment because we could not create a golden truth alignment between H&E and IHC slides. Since the cell-quantification was performed within tiled immune-subtype (immune inflamed, immune excluded, immune desert) regions predicted by a tile-level prediction model from co-aligned H&E images, the pixel-level registration error is statistically insignificant in the cell-quantification within the tiles. Additionally, slides lacking matched anchor pixels were excluded from the integration process. We've provided a figure in the manuscript (please check the attached figure below) outlining the inclusion and exclusion criteria for patients. Please refer to the manuscript for more information.

	CD163	CD20	CD3	CD4	CD45	CD8	ER Alpha	ER Beta	FoxP3	PD-L1	KRT AE1AE3	CD163	CD20	CD3	CD4	CD45	CD8	ER Alpha	ER Beta	FoxP3	PD-L1	KRT AE1AE3
P1	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P2	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o	o
P3	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o	o
P4	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o			o	o
P5	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o	o
P6	o	o	o	o	o	o	o	o	o	o	o											
P7	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o			o	o	o
P8																						
P9	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o	o
P10	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o		o	o	
P11	o	o	o	o	o	o	o	o	o	o	o											
P12	o	o	o	o	o	o	o	o	o	o	o											
P13	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o	o
P14	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o	o
P15	o	o	o	o	o	o	o		o	o	o	o	o	o	o	o	o	o		o	o	o
P16																						
P17	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o	o
P18	o	o	o	o	o	o	o		o	o	o											
P19	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o	o
P20	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o	o
P21																						
P22	o	o	o	o	o	o	o	o	o	o	o											
P23	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o	o
P24	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o	o
P25	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o	o
P26																						
P27	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
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P29	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P30	o	o	o	o	o	o	o	o	o	o	o											
P31	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P32	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P33	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P34																						
P35	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P36	o	o	o	o	o	o	o	o	o	o		o	o	o	o	o	o	o	o	o	o	o
P37	o	o	o	o	o	o	o		o	o	o	o	o	o	o	o	o	o	o	o	o	o
P38	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P39	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P40	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P41	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P42	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P43																						
P44	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P45	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
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P47	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P48	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P49	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P50																						
P51	o	o	o	o	o	o	o	o	o	o	o											
P52	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P53	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o	o	o
P54	o	o	o	o	o	o	o	o	o	o	o											
P55	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P56	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
P57	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o		o	o	o
P58																						
P59	o	o	o	o	o	o	o	o	o	o	o			o		o			o			
P60	o	o	o	o	o	o	o	o	o	o	o			o		o			o			
P61	o	o	o	o	o	o	o		o	o	o											
P62	o	o	o	o	o	o	o	o	o	o	o											
P63	o	o	o	o	o	o	o	o	o	o	o											
P64	o	o	o	o	o	o	o	o	o	o	o			o		o			o			
P65	o	o	o	o	o	o	o	o	o	o	o			o		o			o			
P66																						
P67	o	o	o	o	o	o	o	o	o	o	o	o		o		o			o			
P68	o	o	o	o	o	o	o	o	o	o	o	o		o		o			o			
P69	o	o	o	o	o	o	o	o	o	o	o											
P70	o	o	o	o	o	o	o	o	o	o	o											
P71	o	o	o	o		o	o	o	o	o	o											
P72	o	o	o	o		o	o	o	o	o	o			o		o		o	o			
P73	o	o	o	o		o	o	o	o	o	o											
P74	o	o	o	o	o	o	o	o		o	o			o		o			o			
P75	o	o	o	o	o	o	o	o	o	o	o											
P76																						
P77	o	o	o	o	o	o	o	o	o	o	o			o		o			o			
P78	o	o	o	o	o	o	o	o	o	o	o			o		o			o			
P79	o	o	o	o		o	o	o	o	o	o	o		o		o		o	o			
P80	o	o	o	o	o	o	o	o	o	o	o											
P81	o	o	o	o	o	o	o	o	o	o	o											
P82	o	o	o	o	o	o	o	o	o	o	o	o		o		o		o	o			
P83	o	o	o	o		o	o	o	o	o	o			o		o		o	o			
P84	o	o	o	o		o	o	o	o	o	o			o		o		o	o			
P85	o	o	o	o	o	o	o	o	o	o	o			o		o			o			
P86	o	o	o	o	o	o	o	o	o	o	o			o		o			o			
P87	o	o	o	o	o	o	o		o	o	o			o								
P88	o	o	o	o	o	o	o		o	o	o			o								

4. There is no validation cohort. This is a fundamental limitation to the current study which is critically impactful to its validity. Because there is no validation cohort, the methodological limitations above are quite problematic. A validation cohort would at least provide a simple "but it works" response which would be at least partially sufficient translationally if not scientifically.

   Response: As you pointed out this is the limitation of this study. The validation cohort is hard to obtain with similar curated data.

Followed by a general novelty issue.

1. This is known. There are now dozens of published papers on how TIL infiltration affects survival for OPSCC generally and HPV+OPSCC. It is quite troubling that none of those papers are actually cited or referenced here. Including papers on CD4+ and CD8+; IHC based papers and transcriptomics-based papers.
   Response: We have added significant background literature in the main draft.

2. It has also been shown that TIL infiltration and spatial organization is predictive of OS and DFS in HPV+OPSCC. In multiple independent patient cohorts. Again, troubling that that is neither cited nor referenced here since it precedes the current effort by some time now.
   Response: Thank you for highlighting the addition of corresponding references in the main draft that conducted the TIL-based analysis. It's important to note that while H&E-based TIL analysis is valuable, it inherently lacks the ability to functionally phenotype the TILs. One of the strengths of our method lies in our capability to identify functional subcategories of TILs, such as CD8+ T cells, FoxP3+ T cells, and M2-Macrophages, and explore their correlation with progression-free survival (PFS). By elucidating these functional subcategories, we aim to provide a more comprehensive understanding of the immune landscape and its impact on patient outcomes. Thank you for bringing attention to this aspect, which further emphasizes the significance of our approach.

Added new references:

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REVIEWERS' COMMENTS:

Reviewer #1 (Remarks to the Author):

Thank you for addressing my comments.

Reviewer #2 (Remarks to the Author):

The revised manuscript is responsive to prior critiques. The methodology has been clarified. With the persistent deficit stemming from lack of a validation cohort, I have no outstanding comments or critiques at this point.